EMPOWER presented by isolutions 2025

Deploy Your Microsoft Fabric Architecture with CI/CD in 10 min

Samantha Cruz & Giancarllo Torres

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Agenda

- Introduction
- Key Concepts
    - What is Infrastructure as Code (IaC)
    - Terraform & Modular Design
    - CI/CD with Azure DevOps + Microsoft Fabric
- Live Demo: Deploy in 10 Minutes
- Q&A

About us



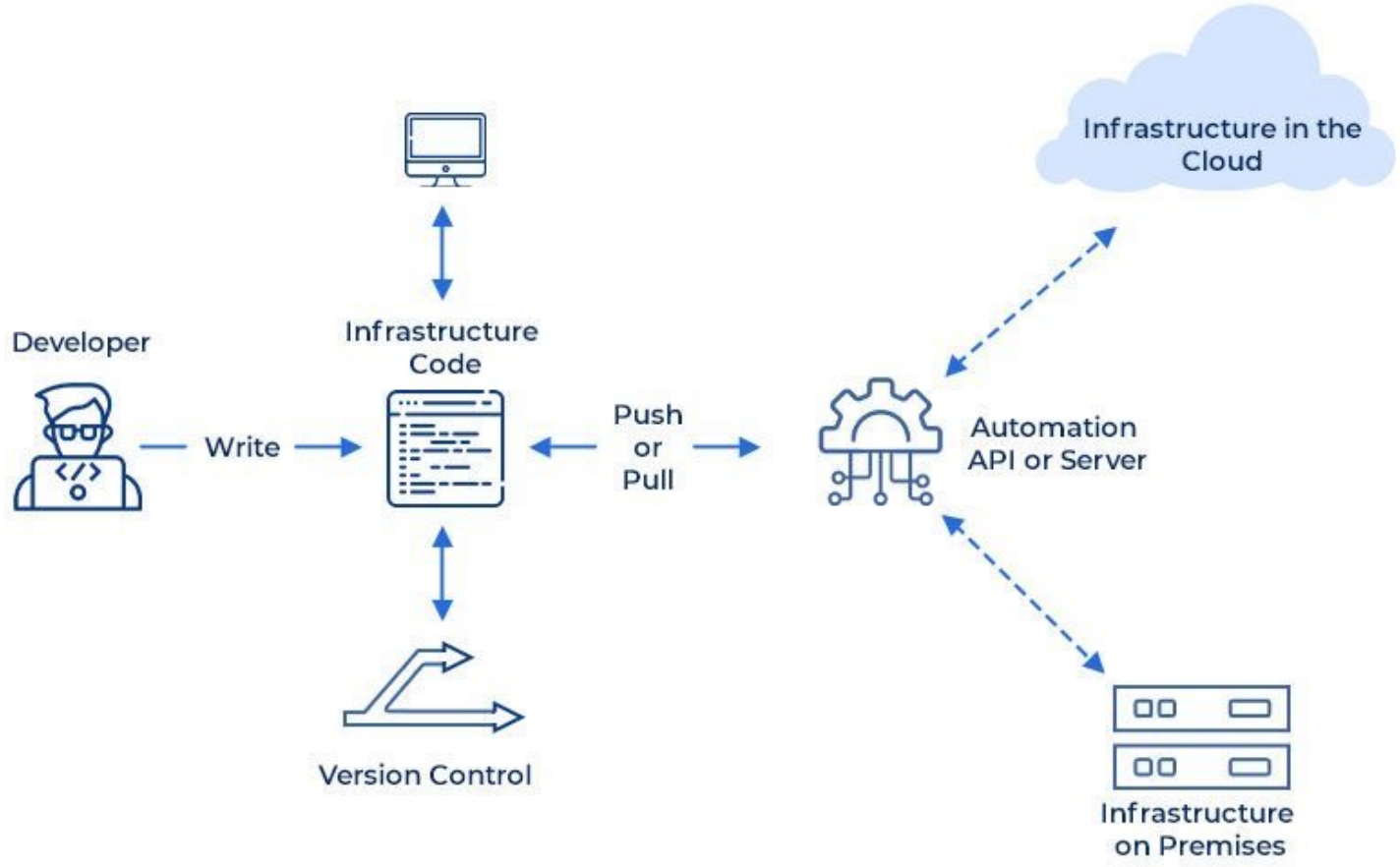
Giancarllo Torres is a Data Engineer and Team Lead at isolutions AG, specialized in ETL and CICD processes across industries, from game-audio analytics at EA to music insights at Universal Music. Passionate about data-driven decision-making, he bridges complex data challenges with actionable business insights.



Samantha Cruz is a Senior Data Engineer at isolutions AG, she builds scalable pipelines, real-time analytics, and end-to-end data platforms using Microsoft Fabric and Databricks with a focus on efficiency and governance. As an educator and mentor, she makes complex data concepts accessible.

Infrastructure as Code (IaC)

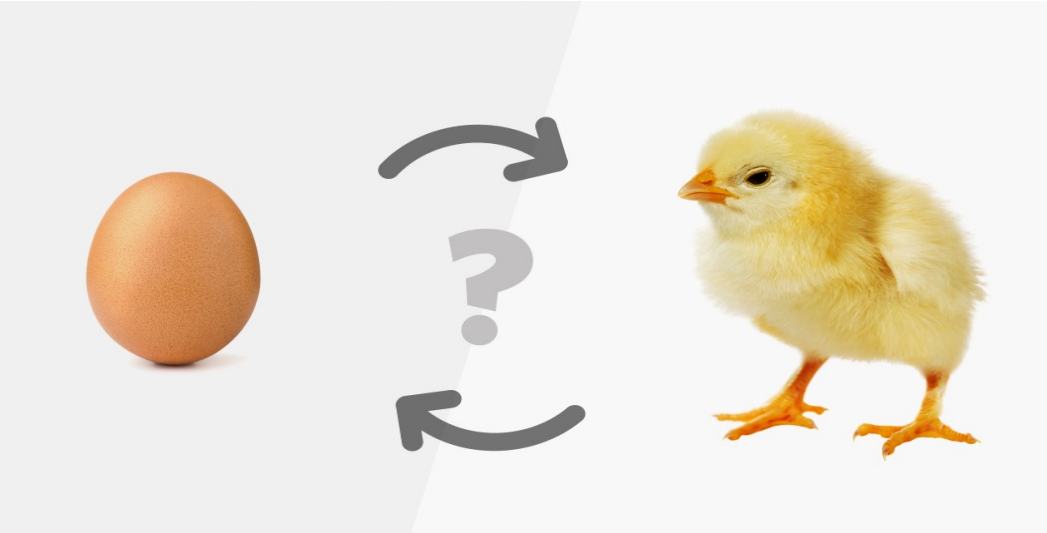
- Define infrastructure using code
- Automate provisioning: no clicks, no consoles
- Version-controlled, repeatable, scalable



Why IaC Matters for Data Projects

- Manual setup ≠ reproducible
- Data teams need reliable environments
- Avoid “it worked on my workspace”

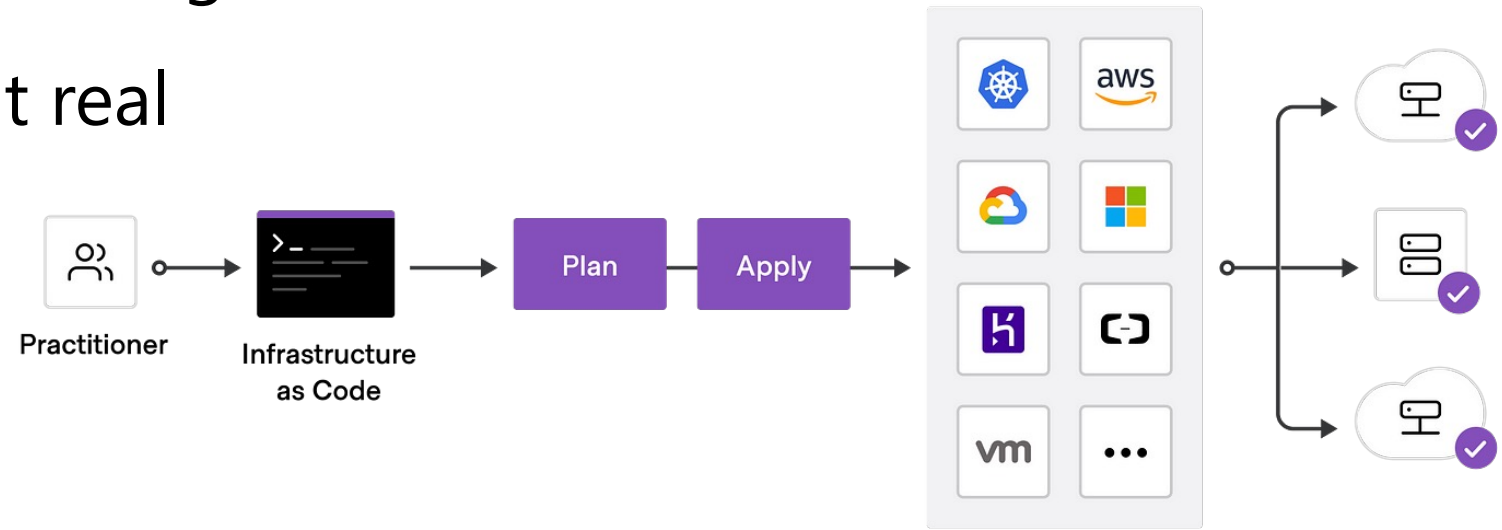
The Chicken and Egg Problem



- Terraform needs a backend to run
- Backend is part of the infrastructure
- How do you deploy the backend… without the backend?

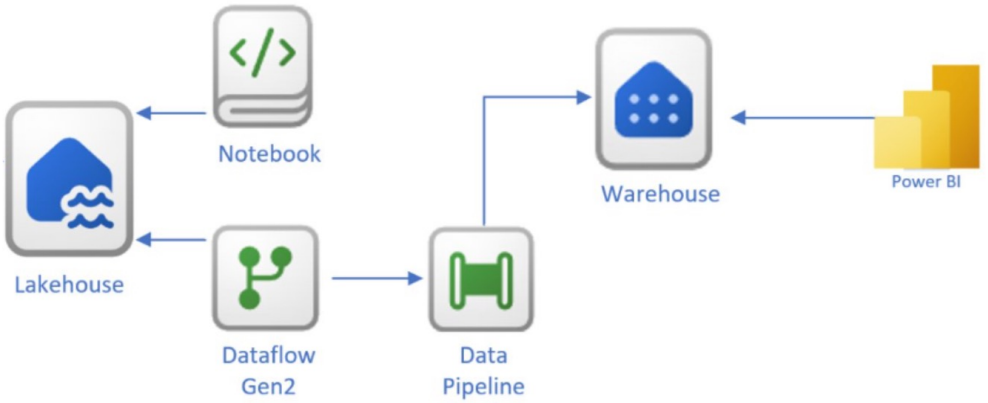
Terraform Basics

- Declarative syntax (.tf files)
- plan → preview changes
- apply → make it real



Modeling Fabric Resources

- Use Terraform Fabric provider
- Define workspaces, lakehouses, manage access, notebooks, etc
- Treat Fabric as any other resource



Modules & Reusability

- Use modules for repeatable patterns
- Parametrize with variables
- One codebase → multiple environments

Azure DevOps Pipelines

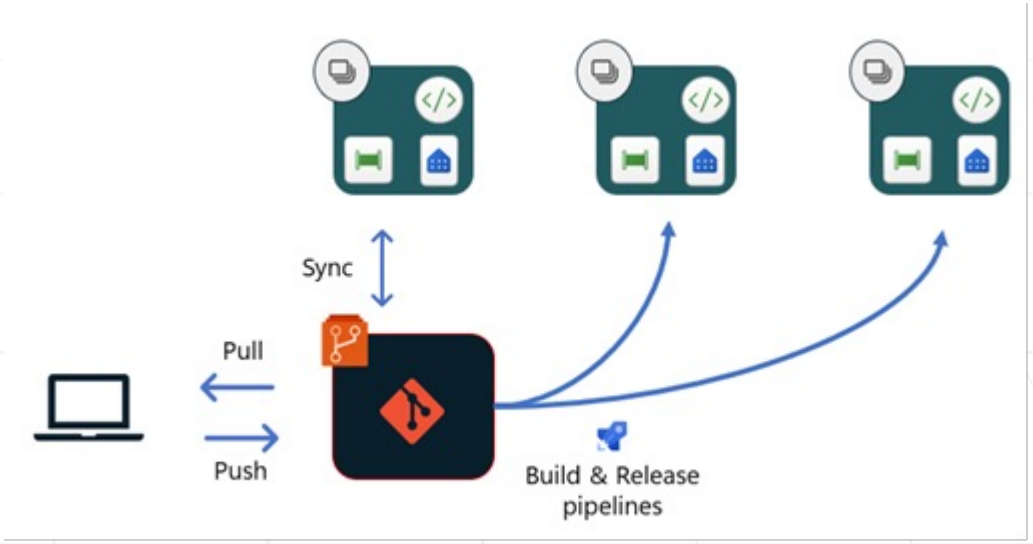
- Multi-stage YAML pipelines
- Steps: validate → plan → apply
- Approval gates for release

Deploying Fabric via CI/CD

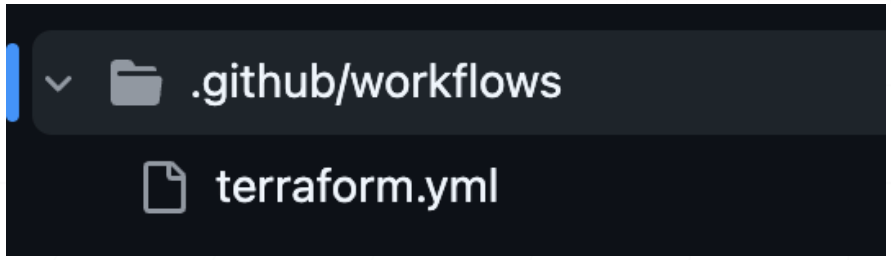
- Inject variables per environment
- Triggered by commits or schedules

Azure Pipelines for Fabric IaC Example

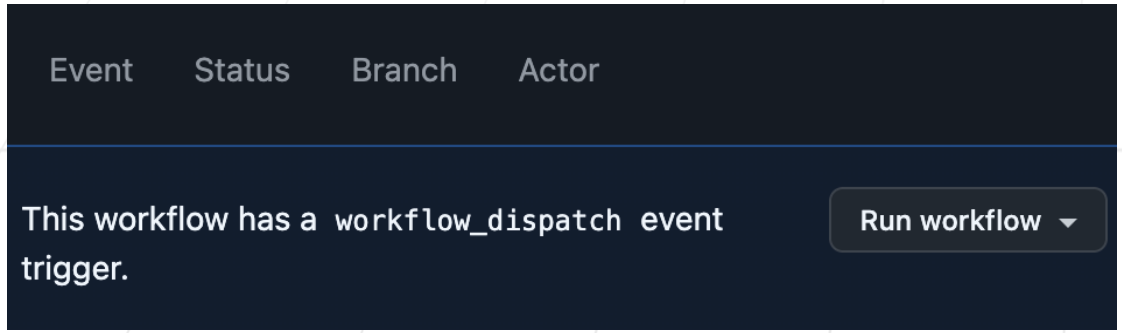
- Inject variables per environment using pipeline variables or library
- Triggered manually or via request



Using Github Actions

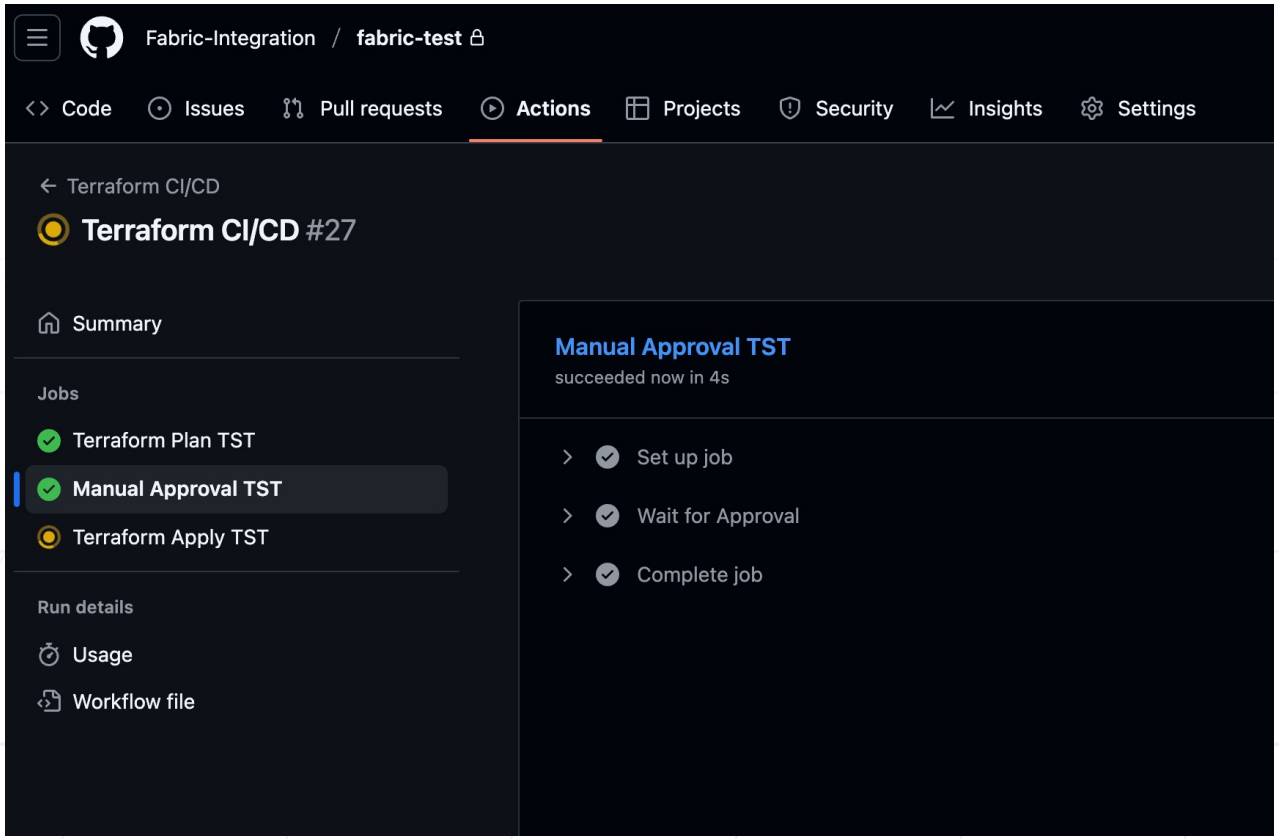


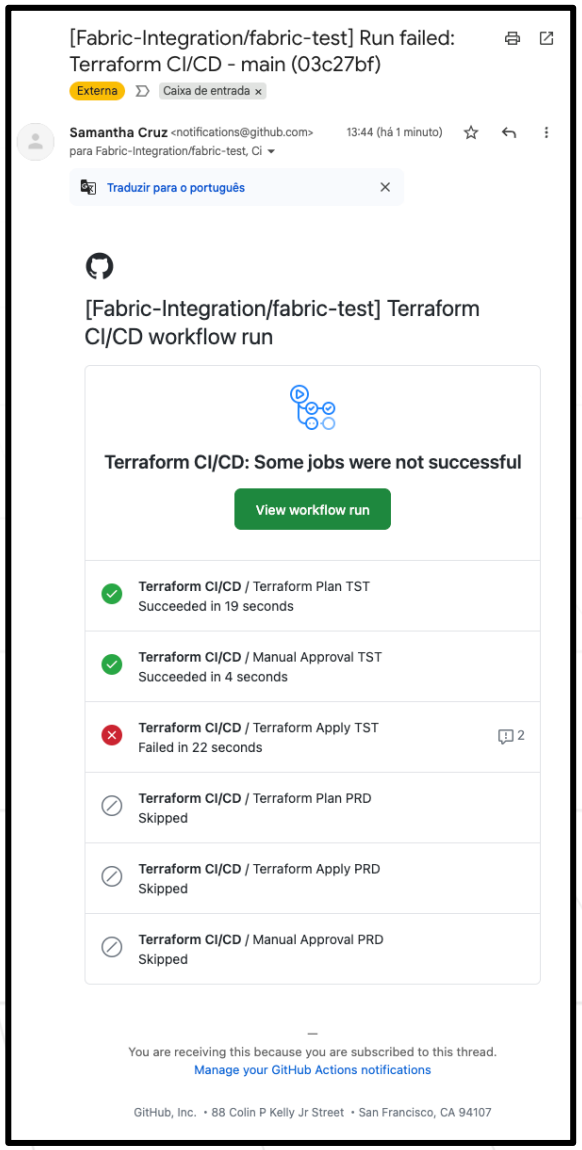




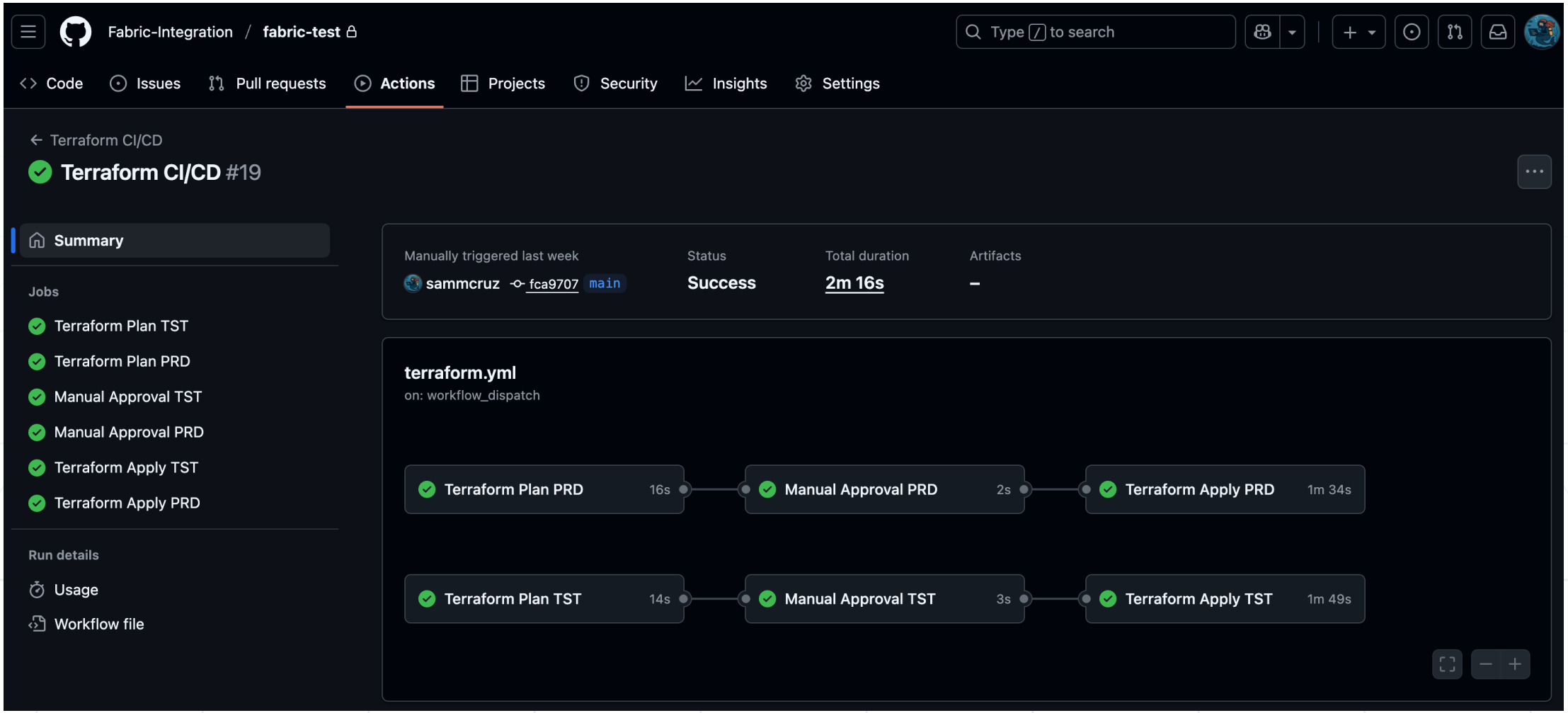
- Store Terraform pipeline as *.github/workflows/terraform.yml*
- Trigger via workflow_dispatch or other GitHub events
- Manual approvals with environment and needs
- Free Tier Limitations:
  - ✅ 2,000 min/month (public repos unlimited)
  - 🚫 No Key Vault or Service Connection UI
  - 🔐 Secrets via GitHub → must be managed manually

Using Github Actions





Using Github Actions



Live Demo: Let's Deploy!

- Requirements:
  - Azure Subscription
  - DevOps Project and Service Connections
  - Fabric Capacity
  - YAML Pipeline and Terraform Code (use our template)

THANK YOU

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